Ayushmaan Puri

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Education

University of California, San Diego — Sep. 2024 - Sep. 2026
B.S. CSE: Computer Engineering — *GPA 3.40/4.0*

Orange Coast College — Aug. 2022 - May 2024
A.S. for Transfer: Computer Science, Mathematics, and Physics – Graduated with Honors — *GPA 3.87/4.0*

Research Experience

Research & Development Engineer | Junkyard Computing — Oct. 2026 – Present
Kastner Research Group – UC San Diego — *Undergraduate Researcher, Mar. – Sep. 2026*

Batteryless Operation and Thermal Testing — *Jun. 2026 – Present*

- Enabled batteryless operation of stripped-down Pixel Fold nodes by wiring regulated DC power to the mainboard and spoofing the battery ID and NTC thermistor signals, found by reverse-engineering the dual Samsung PMIC (s2mpg12/s2mpg13) boot sequence in the felix/gs201 kernel tree; this frees the USB-C port for networking and removes thousands of lithium cells from the rack.
- Ran per-cluster `stress-ng` workloads while logging thermal zones, on-device power rails, and wall-side power from the bench PSU over its USB interface, to physically locate the Tensor G2 CPU/GPU clusters on the mainboard and map hotspots for cooling design and throttling prevention.
- Managing networking and deployment logistics to integrate the phone racks into UCSD's DSMLP data center for course autograding. Working across the thermal/rack, phone teardown, and kernel teams and coordinating with data center staff.

Autograder Cluster — *Mar. 2026 – Jun. 2026*

- Repurposed discarded smartphones as dedicated autograding compute and deployed a 41-phone k3s Kubernetes cluster to replace oversubscribed GPU pods shared across a class of 330 students.
- Modeled real submission timing and runtime distributions from 8 programming assignments using Gaussian Mixture Models with BIC-selected component counts. Built a synthetic workload generator sampling from fitted GMMs to stress test cluster throughput and determine the minimum phone count needed to serve a class without queuing delays.

Undergraduate Researcher | UnifiedSplitting — Jun. 2025 – Feb. 2026
PatLab & UC Scholars Research Program – UC San Diego — *La Jolla, CA*

- Built heterogeneous CPU-GPU pipelines for layer-split CNN inference on unified-memory to reduce CPU idle time.
- Profiled per-layer CPU-GPU speedup ratios across full network architectures; identified a 10x speedup gap between GPU inference of convolutional layers (300–500x) and fully connected layers (30–40x) to determine optimal split points.
- Tested the approach across five architectures (ResNet-18/50, MobileNetV2, EfficientNet-B2, ViT-Tiny) and six simulated edge hardware profiles; compared them to GPU-only baselines. Presented talks at SCCUR 2025 and UCSD SRC 2025.

Summer Research Intern | Active Thermography for Defect Detection — Jun. 2024 – Sep. 2024
SeNSE – Indian Institute of Technology, Delhi — *New Delhi, India*

- Developed a per-pixel signal processing pipeline in Python (OpenCV, NumPy, SciPy) to detect subsurface defects in Carbon and Glass Fibre-Reinforced Polymer (CFRP/GFRP) composites using active thermography.
- Implemented linear detrending, full temporal cross-correlation against a sound reference pixel, main lobe extraction, and FFT phase computation across three experimental datasets (LFM, Active Pulsed, and Flux excitation).
- Demonstrated that the FFT phase of the correlation main lobe produces qualitatively distinct signatures for defective vs. healthy material regions, establishing a new discriminating observable for non-destructive evaluation. Recorded results and findings in a research paper.

Awards and Honors

CRA Outstanding Undergraduate Researcher Honorable Mention | *Computing Research Association* — Jan. 2026

UC Scholar ($5000) | *UC San Diego Undergraduate Research Hub* — May 2025

George Ciarlo Memorial Scholarship ($1000) | *Orange Coast College* — Apr. 2024

Technical Skills

Languages: C/C++, Python, SystemVerilog, Bash, MATLAB, MIPS Assembly
Systems & Infrastructure: Linux, Android/Linux kernel, Kubernetes (k3s), Docker, Git
Parallel Computing & ML: CUDA, OpenCL, PyTorch, scikit-learn, NumPy, SciPy, OpenCV
Hardware & EDA: NVIDIA Jetson Orin, ARM64, ESP32, Cadence Virtuoso, HSPICE